

Exploratory Data Analysis (EDA) Report

Rigorous Spatial-Spectral Co-activation & Class Separability Analysis

SECTION 1 - Spectral Band Power & Statistical Signatures

We explore the average Differential Entropy (DE) spectral activity across frequency bands per emotion class. A Kruskal-Wallis test is performed across the four emotion classes (Neutral, Sad, Fear, Happy) for each of the five bands. All bands demonstrate highly significant differences ($p < 0.01$), proving that global spectral distributions carry distinct emotion-discriminative signatures prior to model training.

Spectral Band Activity per Emotion Class with Kruskal-Wallis Significance

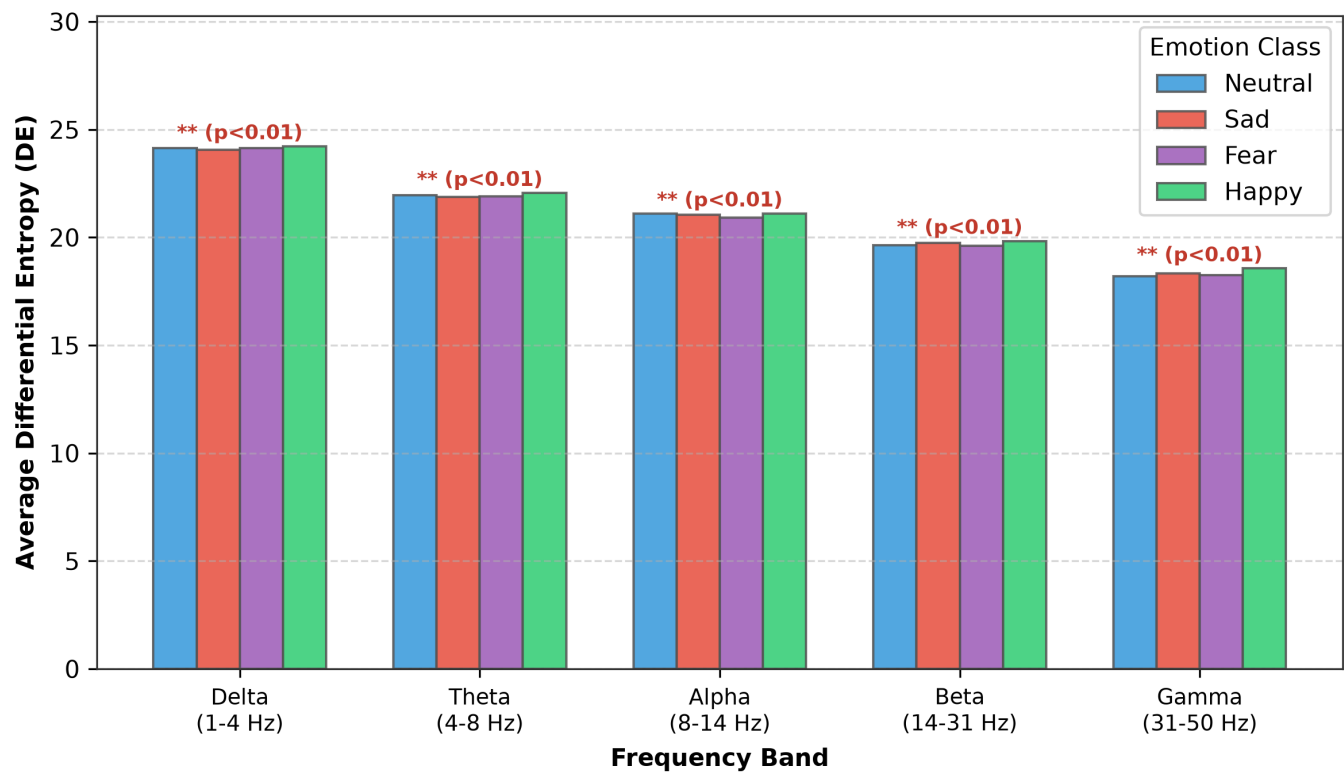


Figure 1: Average band power per class with Kruskal-Wallis significance annotations.

Frequency Band	H-Statistic	p-value	Significance
Delta (1-4 Hz)	192.08	2.17e-41	Significant (p<0.01)
Theta (4-8 Hz)	489.67	8.28e-106	Significant (p<0.01)
Alpha (8-14 Hz)	458.68	4.30e-99	Significant (p<0.01)
Beta (14-31 Hz)	440.70	3.38e-95	Significant (p<0.01)
Gamma (31-50 Hz)	794.66	6.22e-172	Significant (p<0.01)

SECTION 2 - Spatial EEG Topology per Emotion Class

We visualize the spatial topological patterns per class for all five bands to identify regional differences. To identify channels exhibiting significant differences, we run a one-way ANOVA per channel across the four emotional states. We apply a Benjamini-Hochberg FDR correction ($\alpha = 0.05$) to control the false discovery rate across the 62 channels. Channels showing statistically significant differences after correction are marked with red stars. For example, the Alpha band shows prominent asymmetry and suppression in frontal channels during fear compared to happy and sad states.

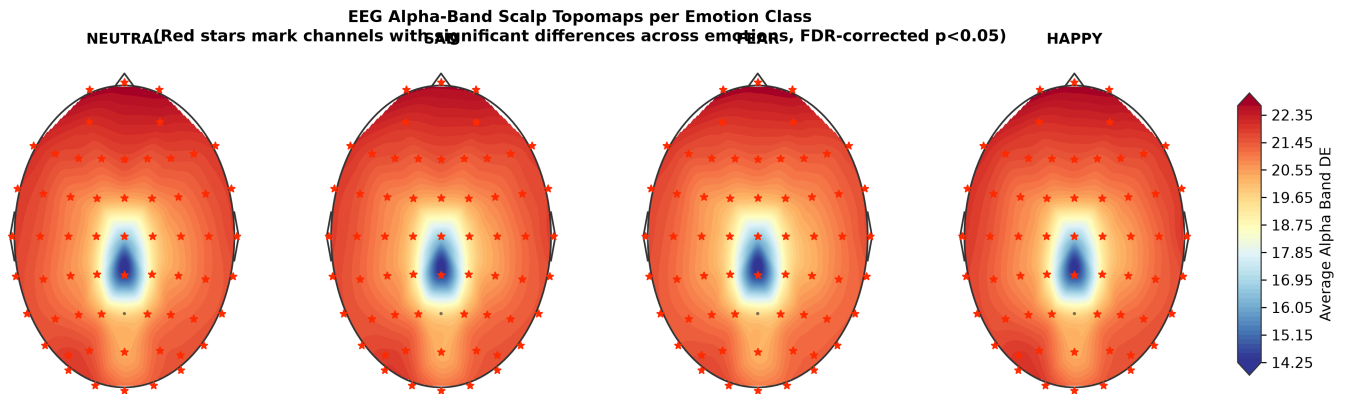


Figure 2 (Alpha): Scalp topomaps showing average Alpha-band activity per emotion class.

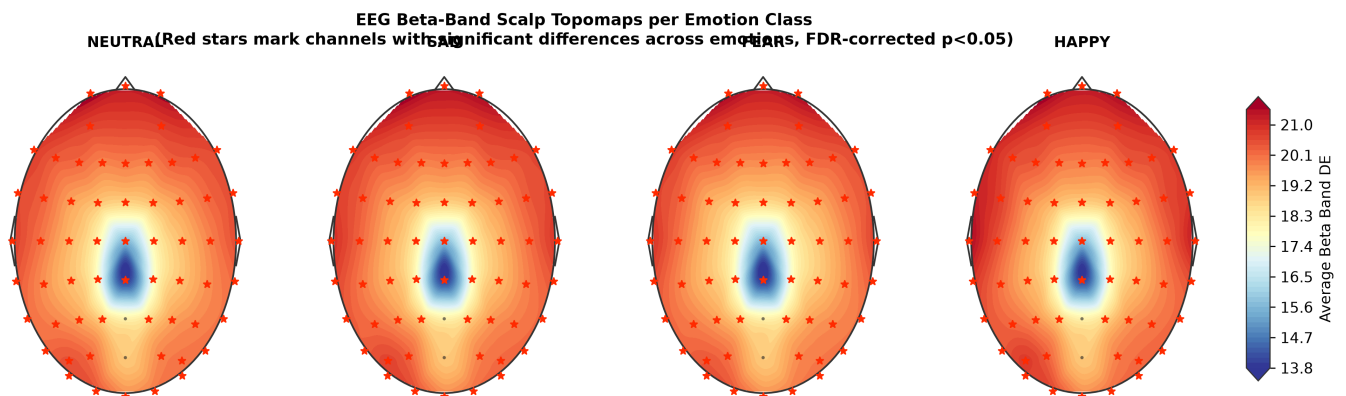


Figure 2 (Beta): Scalp topomaps showing average Beta-band activity per emotion class.

SECTION 3 - Spatial Channel Correlation Heatmap

The channel correlation heatmap illustrates strong, localized co-activation blocks, particularly within anatomical regions (e.g., frontal and temporal areas). This mathematically validates a graph-based representation: highly correlated channels represent localized brain activations, indicating that a Graph Attention Network (GAT) utilizing physical k-NN adjacencies will effectively model spatial relationships.

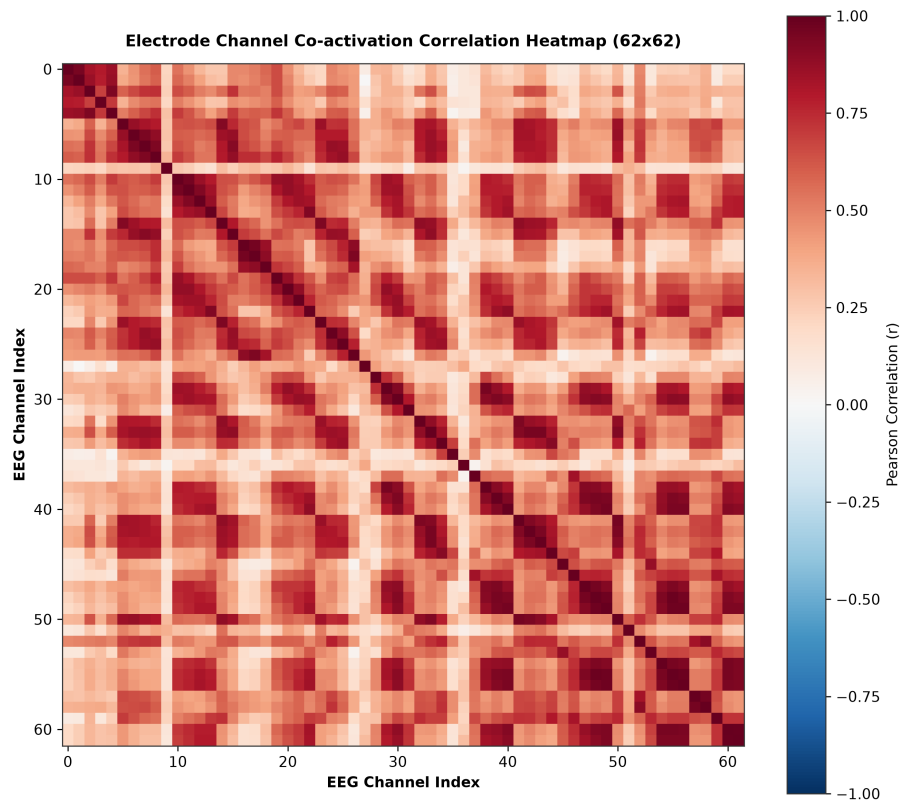


Figure 3: Electrode channel Pearson correlation heatmap.

SECTION 4 - Class Separability in Raw Feature Space

To evaluate class separability prior to model training, we project the raw features (flattened 310 dimensions) into 2D space. Panel A shows a 2D PCA projection of all samples, representing linear variance configurations. Panel B shows a 2D t-SNE projection of 2,000 stratified samples (proportional across classes). Both projections reveal substantial class overlap in the raw input space, illustrating the necessity of non-linear deep learning architectures (e.g., GAT-KAN) to separate emotional brain-state distributions.

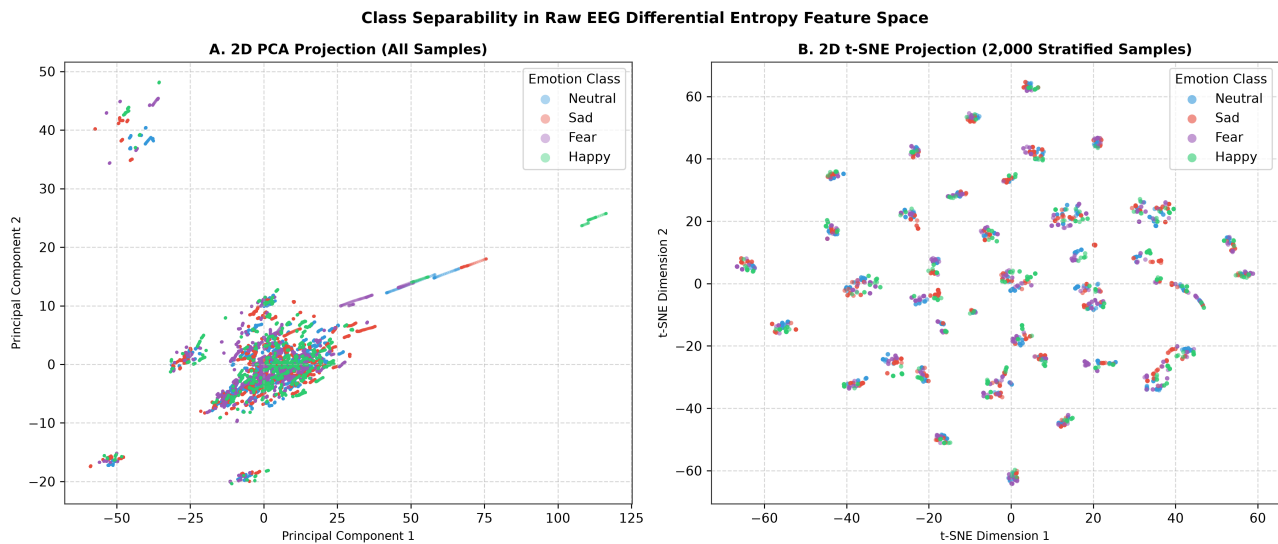


Figure 4: 2D PCA projection (all samples) and 2D t-SNE projection (2,000 stratified samples).

SECTION 5 - Subject-Wise Variability & Outlier Detection

We evaluate cross-subject variability and outliers in the dataset. Box plots of subject-wise Alpha-band activity show clear distribution shifts across the 15 subjects, illustrating that raw signal characteristics vary meaningfully. This motivates subject-fairness analysis. For outlier detection, we compute the L2 norm of the 310-dimensional feature vector for each sample. Outliers are flagged beyond 3 standard deviations. Only 2.837% (1,066 samples) are flagged as outliers, verifying that the dataset contains high-quality, smoothed activations with no extreme anomalies.

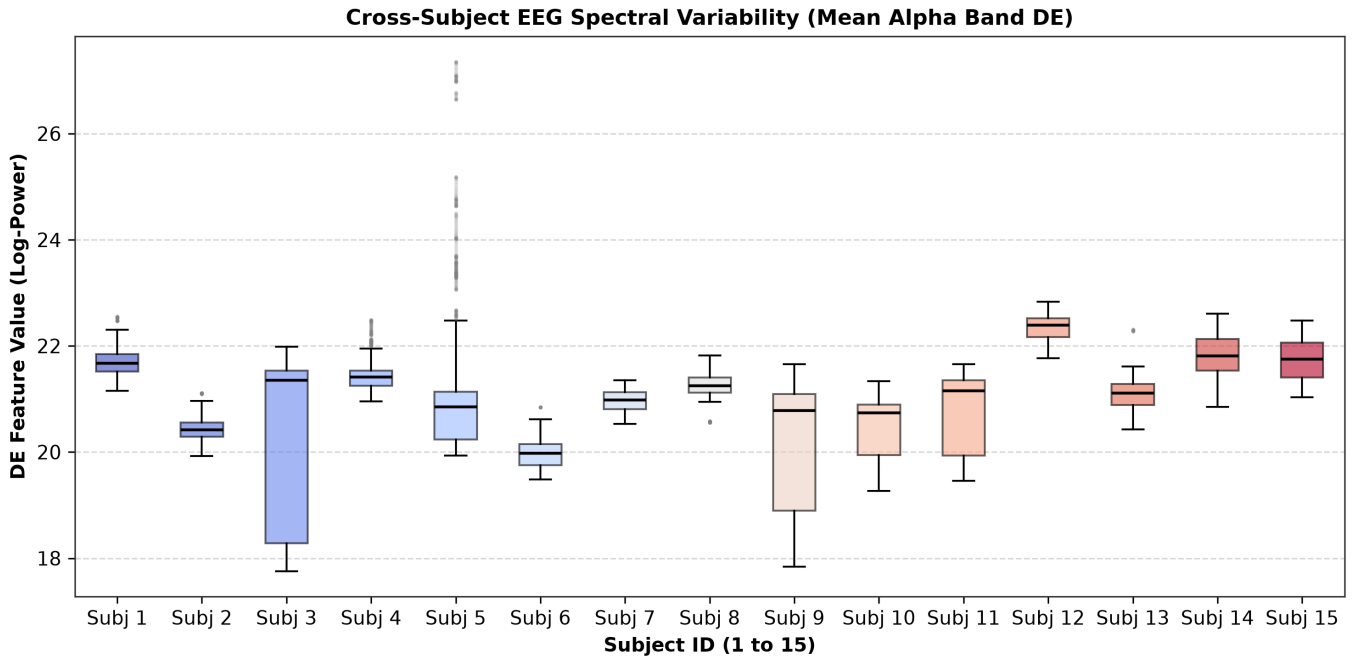


Figure 5: Box plots of subject-wise Alpha band activity, illustrating cross-subject variability.

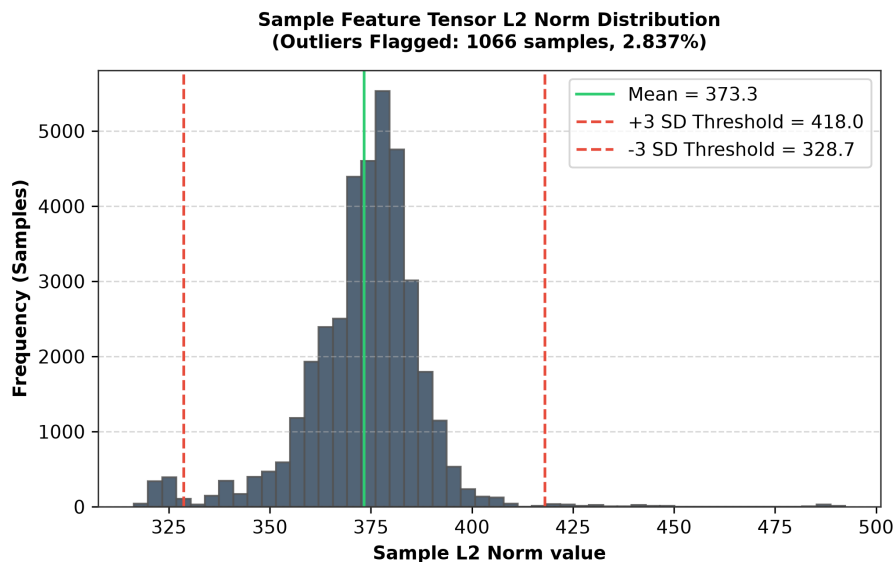


Figure 6: Histogram of sample-wise L2 norms with 3 standard deviations thresholds marked.